Unit 6.3 The dot product p 807 - 813

Note Title 2014/05/22

Definition

$\mathbb{R}^3$

If $\bar{a} = \langle a_1, a_2, a_3 \rangle$ and $\bar{b} = \langle b_1, b_2, b_3 \rangle$ then the dot product of $\bar{a}$ and $\bar{b}$ is given by

$$\bar{a} \cdot \bar{b} = \langle a_1, a_2, a_3 \rangle \cdot \langle b_1, b_2, b_3 \rangle$$
$$= \underbrace{a_1 b_1 + a_2 b_2 + a_3 b_3}_{\text{Number}}$$

Example:

① $\langle -2, 3 \rangle \cdot \langle 4, -5 \rangle = -8 - 15 = -23.$

② $(8\bar{i} - 2\bar{j} + 3\bar{k}) \cdot (2\bar{i} + 4\bar{k}) = \langle 8, -2, 3 \rangle \cdot \langle 2, 0, 4 \rangle$
$= 16 + 0 + 12 = 28$

③ $\langle 2, 3, -1 \rangle \cdot \bar{j} = \langle 2, 3, -1 \rangle \cdot \langle 0, 1, 0 \rangle$
$= 3$

Properties of the dot product p 807

$\bar{a}$, $\bar{b}$ and $\bar{c}$ vectors in $\mathbb{R}^n$ and $c \in \mathbb{R}$

(1) $\bar{a} \cdot \bar{a} = |\bar{a}|^2$

LHS: $\langle a_1, a_2, a_3 \rangle \cdot \langle a_1, a_2, a_3 \rangle$
$= a_1^2 + a_2^2 + a_3^2$

* (2) $\bar{a} \cdot \bar{b} = \bar{b} \cdot \bar{a}$

$|\bar{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

(3) $\bar{a} \cdot (\bar{b} + \bar{c}) = \bar{a} \cdot \bar{b} + \bar{a} \cdot \bar{c}$

$|\bar{a}|^2 = $ LHS:

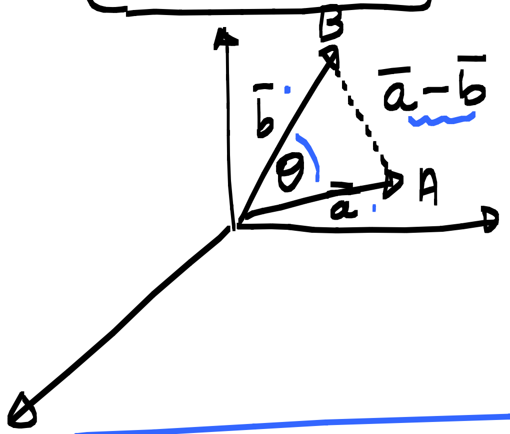
(4) $(c\bar{a})\cdot\bar{b} = c(\bar{a}\cdot\bar{b}) = \bar{a}\cdot c\bar{b}$

(5) $\bar{0}\cdot\bar{a} = 0$ scalar. $\bar{0} = \langle 0,0,0\rangle$ in $\mathbb{R}^3$

Definition

The angle θ between $\bar{a}$ and $\bar{b}$

where $\boxed{0 \le \theta \le \pi}$



Proof not included.

then $\boxed{\bar{a}\cdot\bar{b} = |\bar{a}||\bar{b}|\cos\theta}$ p808

Example:

1. Determine the angle between $\bar{a} = \langle -1, 1\rangle$ and $\bar{b} = \langle 1,1\rangle$

$$\cos\theta = \frac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|} = \frac{-1+1}{\sqrt{2}\cdot\sqrt{2}} = 0$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

Examples:

① Determine the angle between $\bar{a} = \langle -1, 1\rangle$ and $\bar{b} = \langle 1, 1\rangle$.

② Determine the angle between $\bar{a} = \langle 1, -2, 3 \rangle$ and $\bar{b} = \langle -3, 2, 1 \rangle$

$$\cos\theta = \frac{-3-4+3}{\sqrt{14}\sqrt{14}} = \frac{-4}{14} = \frac{-2}{7}$$

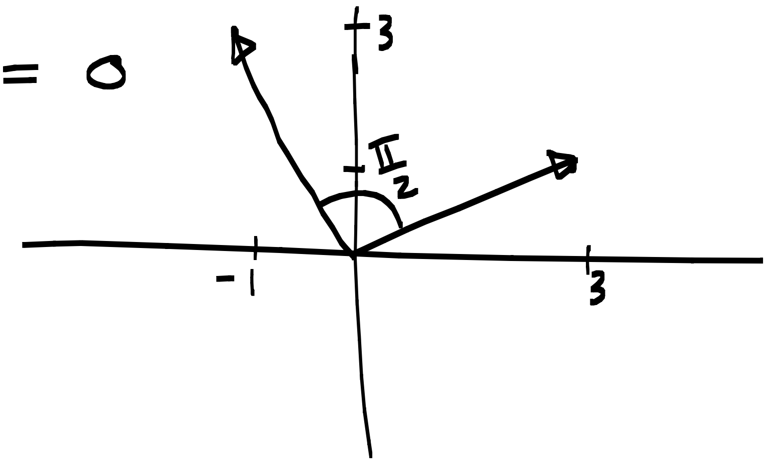
$$\Rightarrow \quad \theta = \arccos\left(-\frac{2}{7}\right)$$

<u>Definition</u> Two non zero vectors $\bar{a}$ and $\bar{b}$ is <u>orthogonal</u> (perpendicular) if the angle between them is $\theta = \frac{\pi}{2}$ that is $\bar{a} \cdot \bar{b} = 0$

<u>Example:</u> ① $\bar{a} = \langle -1, 3 \rangle$ and $\bar{b} = \langle 3, 1 \rangle$

Determine $\bar{a} \cdot \bar{b}$ and sketch the vectors.

$$\bar{a} \cdot \bar{b} = -3 + 3 = 0$$



② Determine k so that $\bar{a} = \langle -1, 3, k \rangle$ and $\bar{b} = \langle 2, k, 3 \rangle$ are orthogonal.

$$\bar{a} \cdot \bar{b} = \langle -1, 3, k \rangle \cdot \langle 2, k, 3 \rangle$$

$$= -2 + 3k + 3k = 0$$

for othogonal.

$$\Rightarrow \quad 6k = 2$$

$$k = \frac{1}{3}$$